

ORGOFLY

UAV Verification, Validation and Test Campaign Handbook

From mission requirements and CAD to subsystem qualification, integration, ground testing, flight-envelope expansion and medical mission validation

Purpose

A practical engineering framework for planning, executing and documenting the OrgoFly tilt-rotor VTOL development test programme. It is designed to help the team ask the right questions, produce traceable evidence and prevent flight testing from becoming unstructured trial-and-error.

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DOCUMENT STATUS

Item	Status / statement
Intended use	Engineering planning guide and reusable test-campaign template for OrgoFly.
System context	Tilt-rotor VTOL fixed-wing UAS; reduced-scale demonstrator followed by larger mission-representative prototype; medical logistics payload.
Regulatory status	Not a certification basis, operational authorisation, flight permit or substitute for competent-authority approval.
Acceptance criteria	Values shown as examples or placeholders must be replaced by project-approved limits derived from requirements, safety analysis, supplier data and the intended operating environment.
Source basis	Synthesised from NASA systems engineering guidance, FAA flight-test guidance, NATO STO UAS flight-test material, EASA UAS design-verification guidance, DoD airworthiness/environmental standards, and WHO/European blood-cold-chain guidance.

Safety boundary

No flight test should begin solely because the aircraft is physically complete. Flight requires a defined configuration, reviewed hazards, closed readiness actions, trained personnel, an authorised test area, approved weather and operating limits, functioning contingency systems, and explicit go/no-go authority. Any team member should be empowered to call an immediate stop for an unsafe condition.

How to use this handbook

- Use Part I to align terminology and understand where testing begins in the development lifecycle.
- Use Part II to create the master Verification and Validation Plan and Requirements Verification Matrix before hardware is complete.
- Use Parts III and IV as a campaign catalogue: select only the campaigns applicable to the current configuration and risk.
- Use Part V to prepare ground and flight-test readiness reviews, test cards, data plans and anomaly closure.
- Use Part VI as the OrgoFly baseline backlog. Tailor the order after confirming the airframe configuration, propulsion architecture, control stack, test site and regulatory pathway.

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1. DIRECT ANSWERS AND CORE CONCEPTS

What is a test campaign?

A test campaign is a coordinated series of verification, validation, qualification, development or acceptance activities conducted against defined objectives on a controlled system configuration. It is larger than one test and smaller than the entire development programme. A campaign normally groups related test points, facilities, configurations and decisions - for example, a propulsion bench campaign, a structural proof campaign, a hardware-in-the-loop campaign, a hover campaign, or a transition-flight campaign.

A campaign is successful only when it produces decision-quality evidence. Merely running the aircraft, collecting video, or accumulating flight hours is not sufficient. The campaign must state what question is being answered, which requirement or hazard is involved, what constitutes pass/fail, what data are required, what configuration was tested, what limitations remain, and what decision follows.

Does testing start before the UAV is built?

Yes - planning starts before CAD is complete.

Verification and validation planning begins with the mission concept, intended users, operating environment, hazards and requirements. CAD is one design representation. Before hardware exists, the team can validate stakeholder needs, inspect requirements, review architecture and interfaces, perform aerodynamic/structural/thermal/electrical analyses, run simulations, assess manufacturability, and define how every requirement will eventually be verified. NASA treats analysis, inspection, demonstration and test as complementary verification methods, while EASA expects the design-verification basis and programme to be agreed early and used to control the evolving UAS design. [1][4]

The correct sequence is not “CAD -> build -> see whether it flies.” It is “mission and hazards -> measurable requirements -> verification strategy -> design and models -> controlled build -> component evidence -> integration evidence -> ground evidence -> progressive flight evidence -> representative mission validation.”

Verification, validation and related terms

Term	Question answered	Typical evidence
Requirements validation	Are the requirements correct, complete, feasible and representative of the real mission?	Stakeholder review, ConOps scenarios, prototypes, simulations, medical-user review, safety review.
Design verification	Does the design or realised product satisfy the specified requirement?	Analysis, inspection, demonstration and controlled test traced to requirement IDs.
Product validation	Does the integrated system perform the intended mission in the intended environment for representative users?	Mission scenarios, representative routes, payload workflow, austere-site operations, user evaluation.
Development test	What do we need to learn or improve?	Exploratory data, model identification, tuning, margin discovery. It may inform design but is not automatically formal compliance evidence.
Qualification test	Can a design withstand the specified lifecycle/environmental conditions with required margin?	Environmental, vibration, structural, endurance or abuse testing on controlled articles.
Acceptance test	Is this particular built article conforming and fit to proceed/deliver?	Incoming inspection, workmanship checks, functional checkout, limited proof or environmental screen.
Regression test	Did a change break previously verified behaviour?	Repeated automated/bench/HIL/ground/flight test subset selected by change-impact analysis.

The hierarchy of test documents

Level	Purpose	Minimum contents
Master V&V / Test Strategy	Explains how the whole programme will generate evidence.	Scope, system breakdown, requirement traceability, campaign tree, facilities, articles, schedule, roles, reviews, data and reporting.
Campaign Plan	Defines one coherent body of testing.	Objectives, requirement/hazard IDs, configuration, test matrix, build-up logic, resources, risks, entry/exit criteria, analysis and deliverables.

Level	Purpose	Minimum contents
Test Procedure	Provides detailed repeatable execution steps.	Setup, equipment, calibration, ordered actions, expected results, tolerances, safety precautions, anomaly instructions and data recording.
Test Card / Run Sheet	Concise real-time execution aid.	Configuration, test point, callouts, limits, aborts, data markers and sign-offs. It must not replace the approved procedure.
Test Report	Captures evidence and conclusions.	As-run configuration, deviations, calibrated data, uncertainty, results, pass/fail, anomalies, corrective actions, limitations and approvals.

How much detail should testing personnel receive?

Use a practical repeatability test: another competent team, unfamiliar with the design history, should be able to set up and execute the test safely without relying on verbal “tribal knowledge.” The level of detail rises with hazard, novelty, irreversibility, complexity, cost of the article, and the difficulty of reproducing the condition.

- Campaign plan: enough detail to explain why the tests are ordered as they are, what risks are being retired and what decision each phase enables.
- Procedure: exact enough to reproduce setup and test conditions, including software versions, wiring/port assignments, sensor orientation, calibration, environmental limits and hold points.
- Test card: short enough for real-time use. It should contain only the actions, callouts, target conditions, limits and immediate contingencies needed during the run.
- Briefing: role assignment, communications, hazards, expected sequence, abort calls, emergency response, data responsibilities and changes since the last test.

2. TEST-CAMPAIGN ARCHITECTURE ACROSS THE FULL LIFECYCLE

Do not create one giant “UAV test campaign.” Divide the programme into linked campaigns, each with its own article, objectives, entry gate and exit gate. The output of one campaign becomes an input to the next.

Phase	Main purpose	Exit product
0. Mission and ConOps	Define users, payloads, routes, operating sites, weather, airspace, crew, maintenance and contingency concepts.	Approved mission scenarios, assumptions and operating limits.
1. Requirements and V&V planning	Convert needs and hazards into measurable requirements; assign verification methods and levels.	Baselined requirements, RVM, V&V plan, initial hazard log.
2. Concept and preliminary design	Compare architectures and close first-order aerodynamic, propulsion, energy, stability, structural and thermal feasibility.	Trade studies, models, design review actions.
3. Detailed design and build readiness	Freeze interfaces, drawings, tolerances, materials/processes, software architecture and test provisions.	CDR/build-readiness approval, controlled design data.
4. Materials and process qualification	Prove materials, joints, bonding, inserts, fasteners, printing/layup/cure processes and inspection methods.	Coupon/joint allowables, process records, acceptance criteria.
5. Component and subsystem bench test	Characterise propulsion, tilt mechanisms, actuators, batteries, sensors, computers, radios and payload equipment.	Performance maps, limits, failure behaviour, accepted components.
6. Software, SIL and HIL	Verify control logic and autonomy against nominal and failed inputs before flight.	Requirements-based results, fault-injection evidence, timing and regression baseline.
7. Structural and propulsion rig tests	Apply representative loads and powered operating conditions to dedicated articles.	Proof/strength/stiffness/modal data, thrust/thermal/vibration maps.
8. Assembly and integration	Verify workmanship, interfaces, configuration, mass properties and end-to-end communication.	Conforming integrated air vehicle, closed interface discrepancies.
9. Ground test	Demonstrate powered integrated operation, safety systems, range, EMI behaviour and mission rehearsal without free flight.	Ground-test report and flight-readiness evidence.

Phase	Main purpose	Exit product
10. Initial hover	Establish controllability, power margin, thermal behaviour, vibration and basic failsafes in the lowest-risk flight condition.	Stable hover envelope and cleared next test block.
11. Transition and fixed-wing envelope	Expand tilt angle, speed, altitude, mass/CG, wind and manoeuvre envelope incrementally.	Transition corridor, handling/performance data, operational limits.
12. Failure/degraded modes	Verify safe response to credible failures using the safest suitable environment: analysis, simulator, HIL, rig, then flight only where justified.	Validated detection, annunciation, containment and recovery logic.
13. Payload and mission validation	Demonstrate representative medical logistics workflow, thermal performance, route, landing, handover and user interaction.	Mission evidence and payload qualification data.
14. Reliability, durability and maintainability	Accumulate controlled cycles/hours and expose the stable configuration to representative lifecycle stresses.	Failure-rate evidence, maintenance intervals, inspections and design improvements.
15. Acceptance and continued configuration control	Apply repeatable checks to every flight article and after significant changes.	Acceptance records, release-to-test status, regression evidence.

Campaign build-up principle

Reduce uncertainty before increasing exposure.

Move from lower-energy, more observable and more controllable environments to higher-energy and less reversible ones: review -> analysis -> simulation -> component bench -> subsystem rig -> SIL/HIL -> integrated ground -> restrained or protected flight -> low-altitude hover -> partial transition -> full transition -> envelope expansion -> representative mission. Do not use flight to discover issues that could have been found safely on a bench.

3. REQUIREMENTS, TRACEABILITY AND DESIGN VERIFICATION BEFORE BUILD

The test programme starts by making requirements testable

A requirement that says “the UAV shall be reliable,” “shall communicate well,” or “shall be structurally sound” is not testable. Rewrite it to define the function, condition, measurable threshold, tolerance, duration, configuration and response to failure. Every test campaign should trace to one or more requirement IDs or hazard-control IDs.

Requirements question bank

- [] What exact stakeholder need or hazard does this requirement address?
- [] Is the subject and required action unambiguous?
- [] Is the requirement measurable, with units, tolerances and a defined reference condition?
- [] Does it state the relevant mass, centre-of-gravity, speed, altitude, weather, payload and power configuration?
- [] Does it cover startup, normal operation, shutdown, maintenance and credible failures?
- [] Are interface timing, data units, coordinate frames, polarity and direction conventions explicit?
- [] Is the verification method identified: analysis, inspection, demonstration or test?
- [] At what level will it be verified: component, subsystem, integrated air vehicle or mission?
- [] What article and facility are required?
- [] What is the objective pass/fail criterion, including allowable uncertainty?
- [] Who owns the requirement and who independently approves closure?
- [] Will a later design change invalidate this evidence?

Minimum Requirements Verification Matrix

Field	What to record
Requirement ID and source	Unique identifier, parent need/hazard, document and revision.
Requirement text	The controlled “shall” statement.
Rationale	Why it exists and what risk it controls.

Field	What to record
Success criteria	Measured value, tolerance, condition and confidence/uncertainty expectation.
Method	Analysis, inspection, demonstration or test; combinations are allowed.
Level and phase	Part, assembly, subsystem, integrated vehicle, ground, flight or mission.
Article/configuration	Serial number, drawing baseline, software/parameter version and payload/CG state.
Facility/equipment	Rig, laboratory, simulator, range and calibrated instrumentation.
Procedure/report	Controlled document IDs and revisions.
Responsible/approver	Execution owner and closure authority.
Status	Planned, ready, passed, failed, conditional, invalidated or retest required.
Evidence and limitations	Data files, plots, report location, deviations and remaining envelope restrictions.

Pre-build verification campaigns

Campaign	Questions and activities	Evidence
ConOps validation	Walk through medical and defence scenarios with operators. Include launch site, loading, route, landing, handover, turnaround, communications, weather, maintenance and emergencies.	Scenario records, user comments, revised requirements.
Architecture/interface review	Does every function have an owner? Are power, data, mechanical, thermal, RF and human interfaces defined? Are units and coordinate frames controlled?	System architecture, ICDs, interface action log.
Aerodynamic/performance analysis	Can the configuration hover, accelerate, transition, cruise, descend and land across mass/CG/environment? What are uncertainties and margins?	Models, CFD/wind-tunnel if needed, performance budget, sensitivity studies.
Stability/control analysis	Is there control authority and actuator bandwidth across modes? What happens at saturation, delay, tilt asymmetry, sensor bias and gust?	Linear/nonlinear analysis, simulation, control-allocation and gain-schedule evidence.
Structural/load analysis	What are limit cases in hover, transition, manoeuvre, gust, landing, transport and handling? Are joints, buckling, stiffness and vibration considered?	Load cases, FEA, margin summary, planned proof/qualification tests.
Power/energy/thermal budget	Does the power system support peak and continuous loads, reserve, thermal rejection and degraded cases?	Load analysis, mission energy model, wiring/protection design, thermal model.
Safety assessment	What failures can cause loss of control, flyaway, fire, uncontrolled descent, payload loss or injury?	FHA/PHA, FMEA/FMECA, fault trees where useful, hazard log and derived requirements.
Manufacturability/testability review	Can the part be built repeatedly and inspected? Are sensors, access points, witness coupons, strain gauges and service connectors designed in?	Manufacturing plan, inspection plan, test-point design, open actions.

Design reviews as test gates

- System Requirements Review: mission, requirements, verification approach and risks are credible.
- Preliminary Design Review: architecture and analyses show a feasible path; major interfaces and test facilities are known.
- Critical Design / Build Readiness Review: drawings, bills of materials, software architecture, processes, inspection points and test provisions are sufficiently controlled to build.

- System Integration Review: components and subsystems have adequate evidence and the integration sequence is safe and defined.
- Test Readiness Review: article, configuration, personnel, procedures, facility, instrumentation, data system, hazards and contingency plans are ready.
- Flight Readiness Review: ground evidence supports flight; airspace/site/weather/crew/emergency provisions are approved and all flight constraints are explicit.

4. SAFETY, RISK AND CAMPAIGN GOVERNANCE

Risk is not a single colour on a slide

A useful hazard record links a hazardous condition to causes, effects, severity, likelihood or qualitative credibility, existing controls, verification evidence, residual risk, responsible owner and approval. The test programme must verify the controls, not merely list them.

UAS hazard question bank

- [] What can produce uncontrolled thrust, loss of thrust, asymmetric thrust or incorrect tilt angle?
- [] What can cause loss of attitude control, navigation, altitude reference, airspeed information or estimator divergence?
- [] What can cause flyaway, departure from the operational volume, loss of C2 or command of the wrong aircraft?
- [] Can a single power, connector, bus, software, sensor or actuator failure defeat all control?
- [] Can a battery, fuel, generator, wiring or motor fault cause fire or thermal propagation?
- [] Can propellers start unexpectedly during assembly, software upload, calibration or recovery?
- [] What structural, aeroelastic or vibration mode could damage the airframe or corrupt sensors?
- [] What happens if a tilt actuator jams, runs away, lags or disagrees with the opposite side?
- [] What happens if GNSS is degraded, spoofed, multipath-affected or lost?
- [] What happens if the GCS display is stale, a mode is misunderstood or a command is delayed?
- [] What hazards are created by the test fixture, tether, tie-down, thrust stand or high-voltage system?
- [] What is the safest termination or recovery action for each phase of flight?

Required campaign roles

Role	Primary authority / responsibility
Test Director	Owns approved execution, sequence, deviations, stop/restart and overall coordination.
Chief / Design Engineer	Confirms technical configuration, expected behaviour, limits and change disposition.
Safety Officer / Range Safety	Owns independent safety assessment, exclusion zones, emergency response and residual-risk concurrence.
Remote Pilot / Test Pilot	Has immediate authority for aircraft control and may terminate a test for any operational concern.
Flight Test Engineer	Defines test points, monitors parameters, coordinates calls and ensures objectives/data are achieved.
Instrumentation/Data Engineer	Owns sensors, calibration, time synchronisation, recording, live limits, file integrity and post-processing.
GCS / Mission Operator	Operates command, telemetry and mission systems; monitors link and mode status.
Maintenance / Airframe Lead	Controls aircraft condition, inspections, servicing and release after configuration changes.
Software/Avionics Lead	Controls binaries, parameters, logs, interface versions and regression status.
Payload/Medical SME	Defines payload conditions, handling, logger placement, acceptance and chain-of-custody requirements.
Quality / Configuration Manager	Ensures article conformity, serialisation, procedure revision, nonconformance and evidence retention.
Observers / Emergency Team	Maintain assigned visual/safety sectors and execute predefined emergency actions.

Entry and exit criteria

Entry criteria - before execution	Exit criteria - before campaign closure
Objective, requirements and hazards are identified.	All planned test points are completed, formally waived or rescheduled with rationale.
Configuration and article identity are frozen and inspected.	Data completeness and quality are confirmed; uncertainty is understood.
Procedure/test cards are approved and rehearsed.	Each requirement is declared pass, fail, conditional or not verified.
Instrumentation is installed, calibrated and time-synchronised.	Anomalies are root-caused or carried as controlled open risks/limitations.
Facility, weather, range and support equipment meet limits.	Configuration changes are incorporated and required regression is complete.
Personnel are trained, briefed and medically/legally fit as applicable.	Reports, plots, raw data and lessons learned are archived under configuration control.
Emergency and abort provisions are operational.	The next phase has explicit clearance conditions and operating limitations.

Standard conduct of a test day

Stage	Actions
1. Configuration confirmation	Check serial numbers, drawings, software hashes/versions, parameters, mass, CG, payload, repairs and open discrepancies.
2. Weather/site/facility confirmation	Record conditions; confirm area, people, communications, emergency assets, lighting and boundaries.
3. Crew brief	Objective, sequence, roles, call signs, target points, limits, aborts, contingencies and changes.
4. Pre-test checkout	Power, telemetry, sensors, logging, clocks, actuator directions, safety devices and independent video.
5. Execute one controlled point at a time	Stabilise, call "ready," mark data, apply input, monitor limits, terminate point, recover and assess.
6. Immediate post-test safeing	Remove energy, disarm propulsion, secure batteries/fuel, preserve logs and inspect for damage/heat/looseness.
7. Debrief and quick-look	Compare expected/observed behaviour, identify anomalies, confirm data quality and decide no-go/conditional/go.
8. Formal analysis and report	Reduce/calibrate data, quantify uncertainty, compare to criteria, close or disposition discrepancies.

5. BUILD, MANUFACTURING AND INTEGRATION CAMPAIGNS

The build itself should be treated as a sequence of acceptance and integration campaigns. Do not wait until final assembly to discover dimensional, structural, wiring or interface problems.

5.1 Incoming material and component acceptance

Questions before parts enter the build

- ☐ Is the supplier, part number, revision, batch/lot and certificate traceable?
- ☐ Does the material match the drawing specification, fibre orientation, alloy/temper, thickness, density or print material?
- ☐ Were storage life, moisture, temperature and shelf-life controls respected for resin, adhesive, prepreg and batteries?
- ☐ Are motors, propellers, servos, bearings, fasteners and electronics genuine and within supplier limits?
- ☐ Are dimensions, mass, balance, electrical continuity and initial functional checks acceptable?
- ☐ Are damaged, altered or substituted parts quarantined and formally dispositioned?

5.2 Materials and process building-block campaign

For composite or bonded structure, use a building-block approach: material coupons -> joints and details -> structural elements -> subcomponents -> full assembly. The objective is to prove both the material behaviour and the repeatability of

the manufacturing process. FAA composite guidance stresses that safety-critical composite structures require control of material/process variability, damage, inspection and repair. [10]

Test level	What to test	Typical outputs
Coupon	Tension, compression, shear, bearing, interlaminar behaviour, environmental conditioning, printed-material orientation and process variability.	Allowables or design values, scatter, failure modes.
Joint/detail	Bonded lap joints, inserts, tube sockets, fastener bearing/bypass, pull-out, peel, local reinforcement, hinge and actuator attachments.	Joint strength/stiffness, acceptable workmanship, inspection indicators.
Element	Spar cap/web, rib bay, skin panel, tube, landing-leg segment, motor-mount arm.	Buckling, local strain, load path and damage initiation.
Subcomponent	Wing half, tail, nacelle/tilt support, fuselage shell, payload-bay frame, landing gear.	Deflection, load distribution, joint performance, proof response.
Integrated structure	Representative limit/proof cases, modal response and critical combined loads.	Structural clearance, inspection baseline and model correlation.

Materials/process question bank

- ☐ Which failure mode is expected: fibre failure, matrix cracking, buckling, adhesive failure, fastener pull-through, insert pull-out or printed-layer separation?
- ☐ Are test specimens representative of the actual process, cure cycle, surface preparation, operator, tooling and geometry?
- ☐ Are witness coupons made with each critical batch or build?
- ☐ How will hidden defects be found: visual inspection, tap test, ultrasound, radiography or another justified method?
- ☐ Are dimensions and fibre/print directions recorded before closure?
- ☐ What proof level is safe and appropriate for the flight article, and what destructive level belongs on a dedicated test article?
- ☐ What post-test inspections are required and what damage is rejectable?
- ☐ How are repairs qualified and how is repaired structure re-verified?

5.3 In-process manufacturing assurance

Build gate	Checks
Before bonding/closing	Surface preparation, fit/gap, adhesive batch, mix ratio, working time, temperature/humidity, fibre orientation, wire routing, embedded hardware, witness coupon and photographs.
During cure/assembly	Cure time/temperature/pressure, clamp/tool condition, fastener type, torque method, locking method, alignment and no trapped foreign objects.
After cure/assembly	Dimensions, mass, visual/NDI result, bond fillet, fastener witness marks, continuity/insulation, freedom of movement and serialisation.
Before next-level integration	Accepted nonconformances, as-built drawing changes, repair records, interface measurements and release signature.

5.4 Electronics and avionics integration campaign

"Do the electronics communicate?" must be decomposed into specific interface and failure questions. A one-time successful boot is not evidence of robust integration.

Area	Nominal tests	Stress/failure tests
Power-up/down	Correct rail voltages, startup order, current, boot time, mode and shutdown.	Brownout, slow ramp, repeated cycling, hot restart, one supply missing, inrush and reset recovery.
Data buses	Message IDs, units, rates, timestamps, ranges, sign conventions and node discovery.	Packet loss, latency/jitter, bus saturation, invalid frames, node reset, cable disconnect, duplicate ID and version mismatch.
Sensors	Orientation, scale, bias, calibration, temperature and reasonableness.	Disconnect, freeze, spike, drift, saturation, disagreement and electromagnetic disturbance.

Area	Nominal tests	Stress/failure tests
Actuators	Direction, travel, rate, position feedback, current and end stops.	Jam, runaway command, feedback mismatch, partial power, thermal load and repeated cycles.
Flight computer	Correct software/parameters, watchdogs, mode state machine and logging.	Process crash, memory/load stress, corrupted input, timing overrun and restart while airborne.
GCS/C2	Command/telemetry, aircraft identity, mode indication and alarms.	Loss/reacquisition, stale data, antenna shadowing, reduced bandwidth, delayed command and wrong operator action.

5.5 First power-on procedure

- Perform continuity, polarity, insulation and short-to-structure checks before energising.
- Use a current-limited bench supply or appropriately protected source for first power where practicable.
- Remove propellers or mechanically isolate propulsion; use an explicit arming inhibit.
- Power one domain at a time, measure current against the predicted load budget and check for heat/odour/noise.
- Confirm processor boot, clocks, bus traffic, sensor orientation and no unexpected actuator movement.
- Exercise each actuator at low authority, verify direction and compare commanded versus measured position.
- Perform controlled power interruption and confirm safe reset/recovery behaviour.
- Capture the wiring baseline, software versions, parameter file and first-power test record.

5.6 System integration / iron-bird campaign

An “iron bird” or systems integration bench uses flight-representative computers, sensors or simulators, actuators, power distribution, C2 and software connected as they will be in the aircraft. It allows end-to-end testing, mode transitions and fault injection before the risks of installed powered testing or flight. The more representative it is, the more valuable it becomes, but its models and differences from the aircraft must be documented and validated. [6][12]

Integration question bank

- [] Does the complete chain work from operator command -> C2 -> flight computer -> control law -> actuator -> position feedback -> telemetry -> display?
- [] Are all clocks synchronised closely enough to reconstruct events?
- [] Do the software state machines allow only valid mode transitions?
- [] Does a reset of one node cause unsafe commands or stale data elsewhere?
- [] Are failure indications timely, unambiguous and visible to the correct crew member?
- [] Does the system remain stable at maximum bus load and logging rate?
- [] Are sensor and actuator models representative enough for the test objective?
- [] Which functions cannot be tested on the bench and must remain open for ground or flight?

6. SUBSYSTEM CAMPAIGN CATALOGUE

Each subsystem campaign should have four outputs: (1) an accepted performance map or capability statement; (2) limits and failure behaviour; (3) an interface/installation release; and (4) evidence that feeds the next integration level.

6.1 Airframe and structural campaign

Test family	What to establish	Key measurements / observations
Mass properties	Weight, CG, inertia estimate and change with payload/fuel/battery.	Calibrated scales, datum, uncertainty, left/right balance, repeatability.
Static stiffness	Deflection and load path under representative wing, tail, nacelle, landing and payload loads.	Load, displacement, strain, joint slip, residual deformation.
Proof/strength	No detrimental damage at the approved proof condition; failure characterisation on dedicated articles where required.	Applied load history, strain, acoustic/visual events, NDI before/after.
Modal / ground vibration	Natural frequencies, mode shapes and sensitivity to mass/configuration.	Accelerometers, excitation force, damping, mode correlation.
Landing/ground loads	Touchdown/drop, tip-over, slope and rough-surface behaviour.	Acceleration, deflection, rebound, local damage and stability.

Test family	What to establish	Key measurements / observations
Durability/fatigue	Repeated representative spectra and inspections.	Cycles, stiffness change, crack/delamination growth, maintenance findings.

Structural design and test questions

- [] What are the critical load cases in hover, transition, cruise, gust, manoeuvre, landing, handling and transport?
- [] Are tilt-nacelle and motor loads applied in all relevant orientations and combinations?
- [] Is stiffness adequate for control accuracy, propeller clearance and avoidance of aeroelastic problems, not only ultimate strength?
- [] Are strain-gauge locations selected from analysis to discriminate load paths and validate the model?
- [] Are test fixtures introducing unrealistic restraint or load concentration?
- [] What damage after proof/load test is acceptable, inspectable and repairable?
- [] What inspection interval is needed for flight test and later operations?

6.2 Propulsion and propeller campaign

Block	Test points	Outputs
Component characterisation	Thrust/torque/RPM versus command, voltage, air density and inflow; efficiency; motor/ESC temperatures.	Performance map and continuous/transient limits.
Propeller integrity/balance	Geometry, mass balance, tracking, overspeed policy, blade/root inspection.	Accepted propellers and inspection/replacement criteria.
Thermal/endurance	Representative hover/cruise cycles, hot-day conditions, repeated starts and cooling.	Temperature margins, derating logic, endurance findings.
Dynamic/vibration	Operating RPM sweep, structural/sensor response, resonance and mounting loads.	No-go RPM bands, mount changes, vibration baseline.
Power-system interaction	Voltage sag, current sharing, brownout, regenerative/transient behaviour, protection.	Peak/continuous power limits and reserve model.
Failure/degraded	Motor/ESC/sensor failure, command loss, partial thrust, incorrect RPM feedback - primarily rig/HIL before flight.	Detection time, control response, safe state and residual capability.

Propulsion questions

- [] Is measured static and forward-flight thrust consistent with the model, including uncertainty and installation losses?
- [] What is the minimum thrust margin at maximum mass, worst temperature/altitude and required control demand?
- [] Can every motor, controller, connector, cable and cooling path sustain the mission duty cycle?
- [] What RPM ranges excite the airframe, tilt mechanism, IMU or payload?
- [] What happens when one unit produces less thrust or responds more slowly?
- [] Does the propeller remain clear of structure through deflection, tilt travel and ground handling?
- [] Are guards, exclusion zones, arming interlocks and emergency shutdown effective?

6.3 Tilt mechanism and control-actuation campaign

Characteristic	Tests	Why it matters
Geometry and travel	Angle calibration, end stops, hard/soft limits, left-right alignment.	Transition schedule and thrust-vector accuracy.
Rate and bandwidth	Step/ramp/chirp under representative aerodynamic/propulsive load.	Control-law assumptions and transition timing.
Backlash/compliance	Reversal, static load, free play and hysteresis.	Limit cycles, asymmetric thrust vector and wear.
Load capacity	Static/dynamic load in hover and forward orientations.	Structural integrity and actuator sizing.
Life/endurance	Representative tilt cycles, lubrication and thermal soak.	Wear, drift, bearing/gear damage and maintenance interval.
Fault response	Jam, lag, runaway command, sensor disagreement, one-sided motion, power loss.	Detection, inhibit, abort and safe recovery strategy.

Tilt-system questions

- [] Is actual angle measured independently from commanded angle?
- [] What is the maximum allowable left-right disagreement at each flight condition?
- [] Can the mechanism hold position under peak thrust and aerodynamic moment without overheating or back-driving?
- [] Is there a mechanically safe end state after loss of power?
- [] Does a sensor fault create a plausible but wrong angle?
- [] Can maintenance or assembly create reversed polarity, wrong zero or different linkage geometry?
- [] What transition abort is possible at every intermediate angle?

6.4 Electrical power and battery campaign

Area	Tests / analyses
Load analysis	All steady, peak, transient and contingency loads by mission phase; margins and shedding priorities.
Distribution/protection	Wire gauge, connector rating, voltage drop, fuses/breakers, isolation, grounding/bonding and single-point failures.
Battery performance	Capacity, internal resistance, voltage sag, temperature, state estimation, ageing and cold/hot conditions.
Abuse and containment	Overcharge/over-discharge controls, short protection, cell imbalance, thermal detection, venting and propagation containment using qualified facilities.
Brownout/recovery	Power interruption, low voltage, bus transfer, processor/servo reset and safe resumption.
Charging/ground handling	Approved charger, storage, inspection, transport, quarantine and fire response.

Electrical questions

- [] Can flight-control power be lost because a propulsion fault collapses the same bus?
- [] Are essential loads isolated or prioritised, and is the transition transient included?
- [] Does state-of-charge estimation remain trustworthy under high current and ageing?
- [] Are connectors keyed, strain-relieved and inaccessible to propeller or moving mechanism damage?
- [] What is the safe response to one failed cell, BMS fault, current-sensor error or temperature-sensor error?
- [] Is battery testing and installation consistent with supplier limits and applicable aviation battery guidance? [11]

6.5 Avionics, flight controls, autonomy and software campaign

Software testing should be requirements-based, configuration-controlled and layered. Certification standards such as DO-178C/DO-254 may not be mandatory for an early prototype, but their core discipline - traceability, reviews, independence appropriate to risk, controlled baselines, problem reporting and change impact - is valuable. [13][14]

Layer	Primary objective	Examples
Unit	Verify individual functions and algorithms.	Math, filters, state transitions, limits, error handling.
Software integration	Verify interfaces and scheduling.	Drivers, messages, timing, memory, logging, startup/shutdown.
SIL	Run flight software or representative logic against aircraft/environment models.	Nominal missions, Monte Carlo, wind/gust, mass/CG, sensor noise.
Processor-in-loop	Check target processor timing and numerical behaviour.	Execution time, overruns, precision, resource use.
HIL	Connect real flight computer/I/O to real-time models and selected hardware.	Actuator/sensor loops, bus faults, latency, C2 and mode transitions.
Ground integrated	Verify installed system, RF/EMI, vibration/thermal effects and full mission rehearsal.	Arm/disarm, actuator signs, lost link, flight termination, payload.
Flight	Validate models and performance in controlled envelope.	Hover, transition, cruise, landing, autonomy, degraded modes.

Flight-control and autonomy questions

- [] Are mode transitions explicit, guarded and reversible, and can two commands conflict?

- [] Does the controller have authority and stability margin across mass, CG, airspeed, tilt angle and actuator delay?
- [] What happens at actuator saturation or rate limiting?
- [] Are sensor validity checks, estimator innovations and disagreement monitors meaningful?
- [] Can a stale value be mistaken for a valid current value?
- [] Are watchdog/reset behaviours safe in hover, transition and cruise?
- [] Does geofence/lost-link logic consider wind, vehicle energy and navigation uncertainty?
- [] Can the pilot always identify the active mode and reason for an automated action?
- [] Are failures injected systematically: bias, drift, dropout, freeze, noise, delay, inversion, packet loss and processor reset?
- [] After a software or parameter change, which regression tests must be repeated?

6.6 Command-and-control, GCS and navigation campaign

Domain	Tests
RF link budget/range	Ground and airborne range, antenna orientation, airframe/rotor shadowing, altitude, terrain and interference.
Latency and integrity	Command and telemetry latency/jitter, loss rate, sequence errors, stale-data detection and time synchronisation.
Loss/reacquisition	Defined lost-link timers/actions, reacquisition, duplicate commands, handover and return-to-home assumptions.
Human interface	Mode awareness, alarm prioritisation, display units, map/altitude reference, workload, acknowledgement and emergency controls.
Navigation sensors	GNSS accuracy/multipath/degradation, IMU bias, magnetometer interference, barometer, airspeed and estimator consistency.
Containment	Geofence accuracy, navigation uncertainty, buffer assumptions, flight-termination independence and test-area boundaries.

C2/navigation questions

- [] What link quality metric is used, and what action occurs before total loss?
- [] Does the aircraft know that a command was received and accepted, and does the operator know its state?
- [] Can antenna placement create a predictable outage during rotor tilt or bank angle?
- [] What happens if home position, altitude datum or map reference is wrong?
- [] Can magnetometer, GNSS or airspeed errors be caused by motors, current, structure or pressure-port installation?
- [] Are navigation and containment systems sufficiently independent for the claimed safety function?

6.7 Environmental, EMC and installation campaign

Environmental testing must be tailored to the actual lifecycle and operating concept. MIL-STD-810 explicitly describes tailoring rather than imposing one universal set of laboratory severities. MIL-STD-461 provides a structured basis for electromagnetic emissions and susceptibility testing. Use these standards as engineering references only after selecting realistic conditions for OrgoFly. [8][9]

Environment	Questions/tests
Temperature/altitude	Cold start, hot soak, hot hover, reduced air density, cooling, sensor drift, battery and payload effects.
Humidity/rain/condensation	Ingress, insulation, corrosion, sensor ports, connector sealing and drying/recovery.
Dust/sand/salt	Motor/bearing/servo wear, cooling blockage, optical sensors, conductive contamination and marine corrosion where relevant.
Vibration/shock	Transport, propulsion, landing and structural modes; connector fretting, IMU corruption and payload exposure.
EMI/EMC	Conducted/radiated emissions and susceptibility; high-current propulsion effects on sensors, radios, GNSS and flight controls.
Solar/UV/storage	Material ageing, enclosure temperature, adhesives, seals and display readability.

6.8 Hybrid propulsion campaign (when introduced)

Hybrid-system question bank

- [] Does the generator/engine meet transient demand during hover, transition and battery recovery?
- [] Can propulsion and avionics remain safe during generator start, stop, surge, misfire or control failure?
- [] How are fuel, exhaust, cooling, vibration, ignition and fire hazards isolated from avionics and medical payload?
- [] What is the power-sharing and load-shedding strategy?
- [] Can the aircraft complete a safe landing after generator failure at each mission phase?
- [] Are fuel quantity, quality, leakage and venting monitored?
- [] Does hybrid-system EMI or vibration corrupt sensors or C2?
- [] What ground endurance hours and inspection intervals are required before flight?

7. INTEGRATED GROUND-TEST CAMPAIGN

Ground testing is the final opportunity to find many installation and integration defects without exposing the aircraft in free flight. It should include both static inspections and powered, mission-representative operation.

Recommended ground-test sequence

ID	Test block	Exit condition
G0	As-built conformity and documentation audit	Drawings/BOM/serials/software/repairs match the released configuration.
G1	Weight, CG and control geometry	Mass/CG within planned test envelope; control and tilt angles calibrated.
G2	Prop-off full functional checkout	All modes, sensors, actuators, arming logic, logging and C2 work safely.
G3	Range/EMI checks	C2/GNSS/sensors remain acceptable with systems and propulsion energised.
G4	Low-power propulsion run	Correct rotation, RPM feedback, current, vibration, cooling and shutdown.
G5	Incremental run-up / tie-down	Thrust, structure, tilt mechanism, thermal behaviour and telemetry through planned ground limits.
G6	Tilt-under-power test	Commanded/measured angle, synchronisation, loads, vibration and abort behaviour.
G7	Failure and contingency rehearsal	Lost link, GCS loss, sensor fault, power reset, emergency stop/FTS and fire response as safely applicable.
G8	Mission dry run	Complete operator workflow and autonomous sequence using restrained/prop-off or safe powered configuration.
G9	Ground endurance / soak	Representative duty cycle to expose heat, connector, software and vibration issues.
G10	Post-ground inspection and FRR package	No damage; anomalies closed; flight limits and first-flight cards approved.

Ground-test readiness questions

- [] Are propellers removed for tests that do not require them?
- [] Are the tie-downs, thrust stand or restraints analysed for the maximum credible load and failure direction?
- [] Could the restraint itself create an unstable dynamic mode or false control response?
- [] Are exclusion zones defined for propeller fragments, battery fire, exhaust and vehicle movement?
- [] Can power be removed immediately and independently of software?
- [] Are motor rotation, control-surface direction and tilt direction verified by two people?
- [] Are live limits displayed for current, voltage, temperature, vibration, tilt disagreement and flight-computer health?
- [] Is the full ground test logged with the same time base and channels intended for flight?
- [] What inspection is required after each run-up level before increasing power?

8. FLIGHT-TEST CAMPAIGN AND ENVELOPE EXPANSION

Flight testing must be task-based and evidence-driven, not a target number of hours. FAA AC 90-89C is written for amateur-built manned aircraft rather than UAS, but its discipline is directly useful: complete a tailored plan before first flight, use test cards, establish controllability and hazardous characteristics, record data, and expand the envelope progressively. [2] NATO STO material similarly treats UAS flight testing as a dedicated discipline involving the aircraft, control station, data links, automation, personnel and risk management. [3]

8.1 Flight readiness review - minimum gate

- Approved test objective, flight card, test area and weather limits.
- As-built configuration and software/parameter baseline confirmed; no unauthorised changes.
- All required ground-test and HIL evidence passed; open anomalies assessed and accepted.
- Mass, CG, payload/ballast, battery/fuel and expected endurance recorded.
- Airworthiness inspection completed, including propellers, fasteners, controls, tilt mechanisms, wiring, antennas and landing system.
- C2, GNSS/navigation, telemetry, onboard logging, video and time synchronisation checked.
- Geofence/containment, lost-link, emergency stop/FTS and recovery provisions verified.
- Crew roles, communication words, knock-it-off authority and emergency plan rehearsed.
- Predicted behaviour, test limits, abort cues and recovery manoeuvre stated for every point.
- Post-flight inspection and decision process agreed before launch.

8.2 Progressive flight blocks

Block	Purpose	Primary clearance objective
F0	Rehearsal without free flight	Crew timing, communications, checklists, data and emergencies.
F1	First lift / short hop	Confirm correct control signs, basic thrust/control margin and safe landing; minimum duration/height needed.
F2	Stable low hover	Attitude, altitude and yaw control; power, vibration, thermal and estimator behaviour.
F3	Hover manoeuvres	Controlled translation, yaw, climb/descent and position hold in low wind.
F4	Hover endurance / mass-CG points	Power/thermal margins and controllability at selected safe test points.
F5	Partial tilt increments	Small angle changes with predefined airspeed/altitude corridor and immediate abort path.
F6	Full transition	Complete transfer from rotor-borne to wing-borne flight only after partial-transition evidence supports it.
F7	Cruise handling and performance	Trim, stability, control response, speed/power, climb/descent and turns.
F8	Transition back / approach	Deceleration, wing-to-rotor lift transfer, altitude control and landing.
F9	Envelope expansion	Mass, CG, speed, altitude, wind/gust, tilt schedules and manoeuvres in controlled increments.
F10	Autonomy and route execution	Waypoint, route, replanning, return, landing accuracy and operator interaction.
F11	Degraded modes	Only after analysis/SIL/HIL/ground evidence; use safe altitude/site and the least hazardous representative failure.
F12	Representative payload mission	End-to-end flight with representative mass, thermal module, route and handover.
F13	Reliability/durability build-up	Stable configuration, repeated representative cycles, planned inspections and failure accounting.

Do not combine novelty on first flight.

The first free flight should not simultaneously introduce an untested software build, maximum mass, extreme CG, high wind, full autonomous mission, full transition and a new payload. Change one major risk variable at a time wherever practical, and preserve a known recovery path.

8.3 Hover campaign

Hover questions and measurements

- ☐ Does total thrust exceed weight with the project-required control and environmental margin?
- ☐ How much command margin remains on every motor during steady hover and manoeuvres?
- ☐ Are attitude/yaw/altitude/position loops stable without persistent oscillation, hunting or saturation?
- ☐ How sensitive is hover to mass, CG, battery voltage, ground effect and wind?
- ☐ Are motor/ESC/battery/actuator temperatures stable over the required duration?
- ☐ What vibration reaches the IMU, structure and payload, and does it corrupt estimation?
- ☐ Do commanded and measured tilt angles remain locked in hover orientation?
- ☐ Does the aircraft drift or rotate after mode changes, sensor resets or command release?
- ☐ Does landing detection, disarm and post-touchdown control behave safely?
- ☐ Do lost-link, geofence and emergency actions produce the expected and safest result for the current altitude?

Parameter group	Minimum useful channels
Vehicle state	Time, position/velocity, altitude references, attitude, body rates, accelerations and estimator health/innovations.
Commands/control	Pilot/autonomy commands, mode/state, control-law outputs, actuator commands/positions and saturation flags.
Propulsion/power	Motor RPM, current, voltage, temperature, ESC state, battery state and total power.
Tilt system	Commanded/measured angle each side, rate, current, disagreement and fault state.
C2/environment	RSSI/SNR/latency/loss, wind, temperature, pressure and ground-effect height.
Payload/structure	Payload temperature, vibration and selected structural strain/acceleration where required.

8.4 Transition campaign - OrgoFly critical test area

Transition is not one test point. It is a corridor in tilt angle, airspeed, angle of attack, pitch attitude, altitude, thrust/power, mass/CG and wind. The campaign must map that corridor and verify both the forward transition and the conversion back to hover.

Transition questions

- ☐ At each tilt angle, what provides lift and what control effectors retain authority?
- ☐ What minimum airspeed/altitude/energy is required before the next tilt increment?
- ☐ How much altitude, pitch and speed deviation occurs during each step?
- ☐ Does the wing approach stall or adverse angle of attack before sufficient speed is gained?
- ☐ Do motors/propellers operate in distorted inflow or produce unexpected pitching/yawing moments?
- ☐ Are tilt actuators synchronised under load, and what disagreement triggers hold/abort?
- ☐ Does control allocation blend smoothly between rotor-borne and aerodynamic controls?
- ☐ Are actuators saturating, rate-limited or opposing one another?
- ☐ What is the abort path at every intermediate angle: return to hover, continue to wing-borne flight or controlled landing?
- ☐ Can transition be completed within battery/power and thermal limits at worst planned mass/environment?
- ☐ What happens under gust, crosswind, one slow motor, sensor bias or delayed tilt response?
- ☐ Is transition-back equally stable, with adequate deceleration and altitude margin?

Example partial-transition test-point build-up

Step	Condition	Action	Stop/abort examples
1	Stable hover at approved altitude, mass/CG and wind.	Confirm all live parameters and reserve.	Any oscillation, saturation, link issue, abnormal temperature/vibration.

Step	Condition	Action	Stop/abort examples
2	Small commanded tilt increment.	Hold angle; measure acceleration, attitude, altitude and control margin.	Tilt disagreement, uncommanded pitch/yaw, altitude loss beyond limit.
3	Recover to hover orientation.	Confirm repeatable reverse response and stable recovery.	Delayed/failed return or increasing oscillation.
4	Repeat at same point.	Establish repeatability and data quality.	Non-repeatable behaviour or unexplained drift.
5	Proceed to next pre-approved increment.	Only after quick-look and Test Director clearance.	Any unresolved anomaly or invalid data.

8.5 Fixed-wing performance and handling

Cruise/envelope questions

- [] What are trim settings, speed stability, control gradients and damping across the approved range?
- [] What is power required versus speed and the resulting endurance/range model?
- [] What climb/descent performance is available at mass, altitude and temperature?
- [] Where are stall onset, control loss or unsafe angle-of-attack behaviours, and how will approach be constrained?
- [] Are there structural, propulsion or control-system vibration boundaries?
- [] How do wind, turbulence and turns affect C2, navigation and payload environment?
- [] What are the safe approach, deceleration and conversion-back procedures?
- [] Are landing accuracy and touchdown loads acceptable at representative sites?

8.6 Failure and degraded-mode testing

Test failures in the safest medium that can answer the question.

Begin with analysis and fault trees, then SIL/Monte Carlo, processor/HIL, component or integrated rigs, and ground tests. Flight-test only failure cases that cannot be adequately verified otherwise and for which the team has a protected test area, altitude/energy margin, independent termination/recovery provisions, and a rehearsed escape path.

Failure class	Preferred early environment	Flight evidence, if justified
Sensor bias/dropout/freeze	SIL/HIL with automated fault injection.	Benign bounded injection at safe condition; confirm detection and recovery.
Actuator lag/saturation	Control analysis, HIL and loaded rig.	Natural/limited condition with ample margin; avoid deliberate destructive jam unless protected.
Motor/thrust loss	Performance/control simulation and propulsion rig.	Only if safe controllability and containment are established; use conservative phase and site.
C2 loss/degradation	Network emulator, HIL, shielded/controlled ground test.	Controlled link interruption with independent safety link/termination as required.
GNSS/navigation degradation	Recorded data, simulation, RF-safe lab methods and HIL.	Controlled operational scenario consistent with legal RF constraints and safety system.
Power brownout/reset	Bench and installed ground test.	Normally not deliberately induced in flight unless specifically designed and independently protected.

9. MEDICAL PAYLOAD AND MISSION VALIDATION

Medical payload testing is a parallel qualification campaign, not an afterthought. The aircraft may meet its flight requirements while the mission fails because temperature, vibration, handling, contamination control, chain of custody or delivery time is unacceptable.

WHO guidance for temperature-sensitive products uses design qualification, operational qualification, performance qualification and route profiling. European blood-quality rules require packaging to maintain product integrity and storage temperature during distribution and transportation. The exact temperature/time limits must come from the medical authority and product specification; do not assume one universal “blood temperature.” [15][16][17]

Stage	Purpose for OrgoFly
Design Qualification (DQ)	Show that the payload-module design, insulation/cooling architecture, sensors, power, mounting, cleanability and capacity are suitable for defined payload classes and route profiles.
Operational Qualification (OQ)	Demonstrate the empty/standard-loaded module under controlled hot/cold ambient profiles, door-open events, power interruption, preconditioning and worst sensor locations.
Performance Qualification (PQ)	Demonstrate repeatable performance with representative payload simulants, actual handling workflow, aircraft vibration and full mission duration.
Route profiling	Measure real ambient, ground-wait, loading, flight, landing and handover conditions on representative routes/seasons; use profiles to define qualification cases.
Clinical/user validation	Have authorised medical stakeholders confirm packaging, labels, chain of custody, logger review, acceptance/rejection and handover workflow.

Medical payload question bank

- ☐ Which exact payload categories are in scope, and what approved time, temperature, shock, vibration, orientation and handling requirements apply to each?
- ☐ What is the maximum and minimum load, packaging arrangement and thermal mass?
- ☐ Where are hot and cold spots, and how many calibrated loggers are required?
- ☐ How is the container preconditioned, loaded, sealed, identified and released?
- ☐ What happens during delayed launch, diversion, landing delay, power loss or failed cooling?
- ☐ How long can the module remain within limits after aircraft power is removed?
- ☐ Does vibration, pressure change or orientation affect the payload or primary container?
- ☐ Can contamination be controlled and the module cleaned without damaging materials/electronics?
- ☐ How are tamper evidence, chain of custody and data integrity maintained?
- ☐ Who decides whether a payload is accepted after an excursion, and what data must be available?
- ☐ Can the loading and handover be performed safely around propellers, batteries and hot surfaces?

Representative mission-validation sequence

- Baseline and calibrate the payload module and loggers.
- Profile the route and ground handling, including worst credible ambient and delays.
- Run chamber/bench OQ cases using representative loads and failure conditions.
- Run vibration and installed-ground tests with the module mounted in the aircraft.
- Conduct short hover flights, then transition/cruise flights with inert representative payload.
- Conduct full-duration mission profiles with representative medical packaging and authorised oversight.
- Analyse temperature, vibration, time, seal/retention, handling and chain-of-custody data.
- Repeat seasonal/worst-case or changed configurations and establish operational release criteria.

10. DATA, INSTRUMENTATION, ANALYSIS AND REPORTING

A test without trustworthy data is often only a demonstration. NASA small-UAS data-collection work recommends capturing vehicle state, motor/actuator commands and condition, C2 information, configuration and contextual reports for off-nominal events. [12]

10.1 Instrumentation plan

Instrumentation planning questions

- ☐ Which parameter proves each acceptance criterion?
- ☐ What range, accuracy, resolution, bandwidth and sample rate are needed?
- ☐ Is the sensor independent of the system under test, or are we only recording what the flight controller already believes?
- ☐ Are all data sources on a common time base, and how is synchronisation verified?
- ☐ What calibration is required before/after the campaign, and how are calibration files linked?
- ☐ Can the recorder survive the event and continue through a power/reset anomaly?
- ☐ Which parameters need live safety limits versus post-test analysis only?
- ☐ How will data loss, clipping, aliasing, dropouts and corrupted files be detected?

[] Are video, audio/calls, ground weather and observer notes synchronised with telemetry?

[] What naming convention and storage/backup process preserves raw data immutably?

Data category	Recommended content
Configuration/context	Article serial, drawing baseline, software hash/version, parameters, mass/CG, payload, battery/fuel, weather, site, crew and procedure revision.
Kinematics/state	Position, velocity, altitude sources, attitude, rates, accelerations, air data and estimator status/innovations.
Control system	Mode/state, commands, setpoints, controller outputs, mixer/control allocation, saturation and limit flags.
Actuation/propulsion	Commanded/measured actuator positions, tilt angles, RPM, current, voltage, torque/thrust where available and temperatures.
C2/navigation	RSSI/SNR, latency, packets, link state, GNSS quality, satellite/solution status, geofence and lost-link state.
Structure/environment	Strain/acceleration/vibration where required, ambient/wind/pressure/temperature and landing loads.
Payload	Temperature/humidity, vibration/shock, door/seal status, power and time stamps for load/handover.
Human/operations	Operator commands, acknowledgements, alarms, callouts, observer notes, deviations and workload/user feedback.

10.2 Acceptance criteria and uncertainty

Define pass/fail before the test. Acceptance criteria should include the measured variable, threshold, duration, condition and treatment of uncertainty. A measured result barely inside a limit may not demonstrate compliance if the measurement uncertainty overlaps the failure region. Conversely, do not invent excessive precision unsupported by the sensors or models.

- State whether a single exceedance, time-averaged exceedance or repeated exceedance constitutes failure.
- Define data validity criteria: stable condition, minimum duration, sample completeness and calibration status.
- Distinguish a test failure from an invalid test caused by wrong setup, configuration or missing data.
- Use confidence intervals/statistics where variability and reliability claims require them; do not treat a handful of flights as proof of fleet reliability.
- Document model uncertainty and update models using test data without retrospectively changing the original acceptance criterion.

10.3 Anomaly process

Step	Required action
Detect and safe	Stop or hold the test; place the article and site in a safe condition.
Preserve evidence	Save raw data, logs, photographs, configuration, parts and witness statements.
Classify	Safety-critical, test-blocking, data-quality, performance discrepancy, workmanship or procedural deviation.
Investigate	Reconstruct timeline; check configuration and test setup; perform root-cause analysis proportionate to risk.
Correct	Design, manufacturing, software, procedure, training or instrumentation action under change control.
Assess impact	Identify requirements and previous evidence invalidated by the change; define regression tests.
Retest and close	Repeat affected points, document results and obtain authorised closure or carry a formal limitation.

10.4 Test report minimum content

- Purpose, scope, requirement/hazard trace and campaign plan reference.
- Test article identity and exact as-run hardware, software, parameters, repairs, mass, CG and payload.
- Facility/site, weather/environment, equipment and calibration records.

- Approved procedure and all deviations or aborted points.
- Raw-data locations, processing methods, filters, corrections and uncertainty.
- Results by test point with plots/tables and comparison to acceptance criteria.
- Pass/fail/conditional/not-tested statement for each objective.
- Anomalies, root causes, corrective actions, retest and residual limitations.
- Lessons learned and recommendation for the next envelope block.
- Reviewer and approval signatures with date and document revision.

11. REUSABLE TEMPLATES AND QUESTION BANKS

11.1 Master Test Campaign Plan template

Section	Required content
1. Purpose and decision	What decision will this campaign enable?
2. Scope/exclusions	Systems, configurations, environments and failure cases included/excluded.
3. References/traceability	Requirement IDs, hazard controls, drawings, ICDs, models and prior reports.
4. Test article/configuration	Serials, hardware/software/parameters, mass/CG/payload and allowed changes.
5. Objectives and success criteria	Measurable criteria, tolerances, uncertainty and data-validity rules.
6. Test matrix/build-up	Ordered blocks and points; rationale for progression.
7. Facilities/equipment	Rigs, site, support equipment, instrumentation, calibration and data system.
8. Roles/training	Named authorities, competence, communications and shifts.
9. Hazards/controls	Hazard analysis, exclusion zones, PPE, energy isolation, aborts and emergency response.
10. Entry/exit gates	Prerequisites, readiness review and closure criteria.
11. Procedures/cards	Controlled document list and approval status.
12. Data/analysis/reporting	Channels, rates, synchronisation, processing, quick-look and final report.
13. Configuration/anomaly control	Deviation authority, change process, nonconformance and regression.
14. Schedule/resources	Dates, dependencies, spares, weather/facility contingencies.

11.2 Test Readiness Review checklist

- ☐ Objectives and pass/fail criteria approved and traceable.
- ☐ Test article received, inspected and identified; configuration list signed.
- ☐ Open discrepancies, waivers and limitations reviewed for test impact.
- ☐ Procedure and cards approved; dry run completed where appropriate.
- ☐ Facility/fixtures/restraints rated and inspected.
- ☐ Instrumentation installed, calibrated, synchronised and channel-checked.
- ☐ Data acquisition, telemetry, storage and backup verified.
- ☐ Hazards, exclusion zones, PPE, fire response and energy isolation approved.
- ☐ Weather/site/airspace and regulatory permissions confirmed for flight.
- ☐ Personnel trained and briefed; roles and decision authority explicit.
- ☐ Emergency/abort/termination/recovery procedures rehearsed.
- ☐ Spares, tools, charging/fuelling and post-test inspection resources available.
- ☐ All readiness actions closed or formally accepted by the correct authority.

11.3 Test procedure / test card template

Header / field	Example content
Identification	Campaign ID, procedure/card ID, revision, date, test article/serial.
Objective / trace	Requirement and hazard IDs; one-sentence purpose.
Configuration	Hardware/software/parameters, mass/CG, payload, battery/fuel, weather.
Entry conditions	Required prior tests, inspection, calibration and stable starting state.
Roles/calls	Test Director, pilot, FTE, safety, data, observers; radio/call words.
Limits	Target, caution, abort and absolute safety limits with responsible monitor.
Procedure	Numbered action; responsible person; expected response; hold point; data marker.
Contingency	Immediate action for each credible abnormal indication.
Data	Channels, sample rate, file name, video/log reference and required annotations.
Post-test	Safeing, inspection, log download, quick-look and release status.
Sign-off	Executed by, deviations, outcome, approval to repeat/proceed.

Example real-time flight-test card skeleton

Seq.	Call / action	Target / data	Abort / recovery
0	Confirm configuration and live limits.	File names, clocks, weather, reserve.	No-go if any mismatch or missing channel.
1	Launch / establish safe condition.	Approved altitude/position/mode.	Land or hold if unstable or link degraded.
2	FTE: "Ready for Point X." Pilot confirms.	Stable precondition for required duration.	Do not start until stable.
3	Mark data; apply specified input.	Magnitude/rate/duration exactly as approved.	Pilot calls "abort" or any member "knock it off."
4	Recover to baseline.	Confirm state, reserve, temperatures and control margin.	Land if recovery is abnormal.
5	Quick-look / repeat decision.	Data quality and repeatability.	No next point with unresolved anomaly.

11.4 Universal questions before any test

- ☐ What exact question are we answering, and why can it not be answered more safely by analysis, simulation or bench test?
- ☐ Which requirement or hazard control will this evidence close?
- ☐ What configuration is being tested, and how do we prove it?
- ☐ What variable are we changing, and what variables must remain controlled?
- ☐ What result do we predict, with what uncertainty?
- ☐ What is pass, fail, invalid and conditional?
- ☐ Which measurements are independent and trustworthy?
- ☐ What can harm personnel, the article, property, payload or airspace users?
- ☐ What is the first indication that the test is diverging?
- ☐ Who watches each critical limit and who has stop authority?
- ☐ What is the immediate abort and safe recovery path?
- ☐ What inspection is needed before repeating or increasing severity?
- ☐ What change would invalidate prior evidence?
- ☐ Where will raw data and the final decision be recorded?

12. ORGOFLY RECOMMENDED TEST-CAMPAIGN BACKLOG

The following is a practical baseline for the current programme. It is intentionally more detailed than a DIANA application. The application can group these into fewer campaign headings, while the engineering team maintains the full internal structure.

ID	Campaign	Primary closure	Gate
OF-TC-001	ConOps and requirements validation	Medical/defence scenarios, users, routes, payload classes, weather, sites, crew and contingencies agreed.	Before design freeze
OF-TC-002	Requirements verification matrix and safety baseline	Every key requirement and hazard control has method, criterion, owner and evidence plan.	Before PDR/CDR
OF-TC-003	Aerodynamic, performance and transition simulation	Hover/cruise/transition feasibility and uncertainty; initial safe corridor.	Before detailed control-law release
OF-TC-004	Structural loads and materials/process qualification	Critical load cases, coupons/joints/elements and manufacturing acceptance.	Before flight-article structure release
OF-TC-005	Tilt mechanism bench qualification	Angle/rate/load/backlash/endurance/fault behaviour.	Before installed powered testing
OF-TC-006	Propulsion and power bench characterisation	Thrust, efficiency, thermal, vibration, endurance and power transients.	Before propulsion installation release
OF-TC-007	Electrical/battery safety and distribution	Load analysis, protection, brownout, BMS, wiring and containment.	Before first integrated power-on
OF-TC-008	Avionics/network interface verification	Message, timing, boot, reset, sensor and actuator communication/fault handling.	Before aircraft integration
OF-TC-009	Flight-control SIL/Monte Carlo	Nominal and uncertain mass/CG/wind/sensor/actuator cases.	Before HIL
OF-TC-010	HIL and fault-injection	Real flight computer, I/O, C2, mode logic, failures and mission rehearsals.	Before powered ground test
OF-TC-011	Airframe manufacturing/acceptance	As-built geometry, workmanship, NDI, fasteners, wiring, mass and CG.	Before system integration
OF-TC-012	Integrated ground functional and EMI/range	End-to-end system, prop-off and powered checks, C2/GNSS/EMI.	Before run-up/hover
OF-TC-013	Tie-down propulsion/tilt/endurance	Installed power and tilt under load; thermal/vibration; emergency stop.	Before first lift
OF-TC-014	Initial hover and landing	Short hops, stable hover, low-speed control, power/thermal/vibration and failsafes.	Before partial transition
OF-TC-015	Hover envelope	Mass/CG, duration, wind, translations, climb/descent and landing accuracy.	Before transition expansion
OF-TC-016	Partial transition	Incremental tilt-angle/airspeed points with recovery to hover.	Before full transition
OF-TC-017	Full transition and conversion back	Complete forward and reverse conversion; initial corridor and abort procedures.	Before cruise envelope
OF-TC-018	Cruise performance and handling	Trim, stability, power-speed, climb/descent, turns and approach.	Before representative route
OF-TC-019	Envelope and degraded modes	Selected mass/CG/wind/altitude; rigorously justified failures after HIL.	Before mission validation
OF-TC-020	Payload DQ/OQ/PQ and route profile	Thermal, vibration, retention, handling, power interruption and logger qualification.	Parallel; before medical mission
OF-TC-021	End-to-end civilian medical mission	Representative pickup, flight, landing, handover and return with payload evidence.	Mission demonstration
OF-TC-022	Defence/austere-site mission	Small-team deployment, alternative payload, temporary site and communications/navigation robustness.	Dual-use demonstration

ID	Campaign	Primary closure	Gate
OF-TC-023	Reliability/durability/maintainability	Stable configuration hours/cycles, inspections, failures, repair and turnaround.	Before operational maturity claim
OF-TC-024	Full-scale/hybrid regression programme	Repeat affected structural, propulsion, control, payload and flight campaigns for larger/hybrid configuration.	Before full-scale mission demo

Highest-priority items for OrgoFly now

- Freeze the current 1:4-scale configuration and create its RVM; separate measured prototype capability from full-scale design targets.
- Create the hazard log and derive explicit requirements for tilt asymmetry/jam, loss of C2, power loss, motor failure, sensor faults, geofence/containment and battery fire.
- Instrument commanded and measured tilt angle, actuator current, motor RPM/current/temperature, flight modes, estimator health and link metrics before attempting serious transition testing.
- Complete loaded tilt-mechanism and propulsion bench campaigns before using flight to discover mechanism or power limits.
- Build a real-time HIL/systems bench that uses the actual flight computer, software, GCS and relevant I/O; automate regression and fault injection.
- Define a partial-transition matrix with one tilt increment at a time, explicit altitude/airspeed prerequisites and a recovery path at every point.
- Run payload thermal DQ/OQ work in parallel rather than waiting for the full-scale aircraft.
- Use a formal change-impact process: airframe geometry, control gains, propellers, motors, batteries, tilt linkage, software and payload changes can invalidate earlier evidence.

Information needed to tailor this handbook into executable OrgoFly plans

Input	Why it changes the campaign
Current prototype mass, dimensions, rotor count/layout and tilt kinematics	Determines load cases, thrust margin, control allocation and safe test sequence.
Materials and manufacturing processes	Determines coupon/joint tests, NDI, proof loads and acceptance records.
Motor/propeller/ESC/battery specifications and measured bench data	Determines power limits, thermal monitoring and first-flight endurance.
Flight controller, autopilot software and actuator network	Determines SIL/HIL architecture, logging, fault injection and software assurance.
Manual, assisted and autonomous modes	Determines pilot training, mode-transition tests and human-interface risks.
Existing completed flights and anomalies	Prevents claiming tests not performed and identifies the next justified envelope block.
Test site, airspace authorisation and safety equipment	Determines containment, observers, emergency response and permissible failure testing.
Full-scale MTOW, hybrid architecture and payload-module concept	Determines which scale results transfer and which campaigns must be repeated.
Medical partner payload specifications	Defines legitimate thermal/time/vibration and chain-of-custody criteria.

13. SOURCE NOTES AND REFERENCES

This handbook combines generally applicable systems-engineering and flight-test practice with OrgoFly-specific tailoring. Standards written for manned aircraft, military airworthiness, space hardware or certification programmes are used as engineering references, not automatically imposed requirements. The eventual compliance basis must be agreed with the relevant aviation authority and, for defence use, the applicable military airworthiness authority.

Reference	Use / link
[1] NASA	NASA Systems Engineering Handbook, NASA/SP-2016-6105 Rev 2. Product verification/validation, requirement quality, verification methods, RVM and lifecycle reviews. https://explorers.larc.nasa.gov/2023ESE/pdf_files/NASA04.%20SysEng%20Handbook%20%28NASA_SP-2016-6105%20Rev%20%29.pdf
[2] FAA	AC 90-89C, Amateur-Built Aircraft and Ultralight Flight Testing Handbook, 14 February 2023. Used for task-based flight-test planning discipline and test-card examples; not a UAS certification standard. https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_90-89C.pdf
[3] NATO STO	Flight Test Methods for Unmanned Aircraft, STO-MP-SCI-328-04, and associated SCI-328 proceedings. https://publications.sto.nato.int/publications/STO%20Meeting%20Proceedings/STO-MP-SCI-328/MP-SCI-328-04.pdf
[4] EASA	Guidelines on Design Verification for UAS Operated in the Specific Category, Issue 3, 26 September 2023. https://www.easa.europa.eu/en/downloads/126318/en
[5] EASA	Functional Test Based Means of Compliance with SC Light-UAS, consultation/proposed material, 2022. Use as an illustrative methodology and verify current status before regulatory reliance. https://www.easa.europa.eu/sites/default/files/dfu/ftb_moc_to_sc_light-uas_0.pdf
[6] EASA	SC-VTOL / UAS means-of-compliance material on qualifying simulation benches and test rigs. Verify the current revision and applicability on the EASA Means of Compliance pages.
[7] DoD / DLA	MIL-HDBK-516, Airworthiness Certification Criteria. Applies foundational criteria to manned and unmanned fixed-/rotary-wing air systems and is tailored by the responsible authority. https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=212162
[8] DoD / DLA	MIL-STD-810, Environmental Engineering Considerations and Laboratory Tests. Emphasises lifecycle environmental tailoring rather than a universal test recipe. https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=35978
[9] DoD / DLA	MIL-STD-461, Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment. https://quicksearch.dla.mil/qsdocdetails.aspx?ident_number=35789
[10] FAA	AC 20-107B with Change 1, Composite Aircraft Structure. https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_20-107B_with_change_1.pdf
[11] FAA	AC 20-184, Guidance on Testing and Installation of Rechargeable Lithium Battery and Battery Systems on Aircraft. https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_20-184_Final_proof.pdf
[12] NASA	Initial Approach to Collect Small Unmanned Aircraft System Off-Nominal Operational Situations Data, NASA NTRS 20180004318, 2018. https://ntrs.nasa.gov/api/citations/20180004318/downloads/20180004318.pdf
[13] FAA	AC 20-115D, Airborne Software Development Assurance Using EUROCAE ED-12 and RTCA DO-178. https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_20-115D.pdf
[14] FAA	AC 20-152A, Development Assurance for Airborne Electronic Hardware. https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_20-152A.pdf
[15] WHO	Transport Route Profiling Qualification, Technical Supplement to WHO TRS 961 Annex 9. https://cdn.who.int/media/docs/default-source/medicines/norms-and-standards/guidelines/distribution/trs961-annex9-suppl4.pdf
[16] WHO	Model Guidance for the Storage and Transport of Time- and Temperature-Sensitive Pharmaceutical Products, WHO TRS 961 Annex 9, plus shipping-container and monitoring supplements. https://www.who.int/docs/default-source/medicines/norms-and-standards/guidelines/distribution/trs961-annex9-modelguidanceforstoragetransport.pdf
[17] European Commission / EDQM	Commission Directive 2005/62/EC: packaging shall maintain integrity and storage temperature of blood or blood components during distribution and transportation. https://www.edqm.eu/documents/52006/66281/directive-2005-62-ec-on-quality-systems-for-blood-establishments-september-2005.pdf/

Important source-status notes

- EASA and national UAS rules, means of compliance and operational-authorisation processes evolve. Confirm the current issue and applicability before using any document as a formal compliance basis.
- FAA AC 90-89C is not written for UAS; it is cited for disciplined flight-test planning, documentation and progressive envelope expansion.
- MIL-HDBK-516, MIL-STD-810 and MIL-STD-461 should be tailored by qualified authorities/engineers. Testing to an arbitrary military-standard method without a mission-derived requirement can waste resources or create misleading evidence.
- WHO and blood-establishment requirements are payload-specific. The medical partner must define approved product limits and release/rejection procedures.

Bottom line

A good test programme does not ask only “does it fly?” It asks whether the right system has been built, whether every critical requirement and hazard control has objective evidence, whether the evidence applies to the exact configuration, whether the limits are understood, and whether the integrated aircraft can perform a representative medical mission safely and repeatedly.